A Systematic Review of Privacy Policy Literature

YOUSRA JAVED, Illinois State University, Normal, United States
AYESHA SAJID, National University of Sciences and Technology, Islamabad, Pakistan

An organization's privacy policy states how it collects, stores, processes, and shares its users' personal information. The growing number of data protection laws and regulations, as well as the numerous sectors where the organizations are collecting user information, has led to the investigation of privacy policies with regards to their accessibility, readability, completeness, comparison with organization's actual data practices, use of machine learning/natural language processing for automated analysis, and comprehension/perception/concerns of end-users via summarization/visualization tools and user studies. However, there is limited work on systematically reviewing the existing research on this topic. We address this gap by conducting a systematic review of the existing privacy policy literature. To this end, we compiled and analyzed 202 papers (published till 31st December, 2023) that investigated privacy policies. Our work advances the field of privacy policies by summarizing the analysis techniques that have been used to study them, the data protection laws/regulations explored, and the sectors to which these policies pertain. We provide actionable insights for organizations to achieve better end-user privacy.

CCS Concepts: • **Social and professional topics** → **Privacy policies**; • **Security and privacy** → **Usability in security and privacy**; **Privacy protections;**

Additional Key Words and Phrases: Privacy policy, systematic literature review, survey, data protection, personal information

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1 Introduction

A privacy policy, also referred to as a privacy statement, is a legal document that describes how an organization collects, processes, and shares their users' personal information, how the integrity and confidentiality of this information is ensured, and what rights the users have for controlling access to this data [1].

There are various data protection laws and regulations that create requirements for the privacy of users' personal data. For instance, the enactment of the **General Data Protection Regulation (GDPR)** in May 2018 [3] led to a need for businesses inside the **European Union (EU)**, as well as those outside the EU, who had customers in the EU, to comply with GDPR and have a privacy policy. Similarly, the number of sectors where the organizations are collecting user information has increased exponentially with the rise of mobile, IoT devices and cloud service providers. This

Authors' Contact Information: Yousra Javed, Illinois State University, Normal, Illinois, United States; e-mail: yjaved@ilstu.edu; Ayesha Sajid, National University of Sciences and Technology, Islamabad, Pakistan; e-mail: asajid.msis19seecs@seecs.edu.pk.



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has led to an extensive investigation of privacy policies with respect to their accessibility, readability, completeness, comparison with organization's actual data practices, use of machine learning/natural language processing for automated analysis, and comprehension/perception/concerns of end-users via summarization/visualization tools and user studies.

Numerous literature reviews have been conducted on specific aspects of privacy policies, such as those focused on the healthcare sector [29], automated analysis using machine learning and NLP approaches [64], language and presentation format [107], and visualization of user information [34]. However, to the best of our knowledge, there is no existing comprehensive systematic literature review that covers the breadth of published privacy policy research, summarizing the existing work in terms of the various methods used for analysis, as well as the investigated data regulations/laws and the explored sectors. Such review can aid in understanding the breadth of research on privacy policies and identify gaps [131, 144], be used to develop conceptual background for guiding future research and policy development, understand compliance requirements, as well as provide actionable insights for organizations to enhance their privacy practices.

We contribute to the existing literature on privacy policies by conducting a systematic review of 202 full-length peer-reviewed research articles focused on privacy policies, published till 31st December, 2023. We consolidate the analysis methods applied to privacy statements, examine the relevant data protection laws and regulations, and identify the sectors impacted by these policies. Additionally, we offer practical recommendations for organizations to enhance user privacy. Our article investigates the following research questions:

(1) What methods have been used to analyze privacy policies?
(2) What data protection laws and regulations (e.g., GDPR) have been used to investigate these policies?
(3) Which sectors have been investigated, e.g., finance, e-commerce, health?

The remainder of this article is organized as follows: Section 2 provides a background on privacy policies, data protection laws, and regulations and discusses related work in this domain. Section 3 describes our data collection methodology. Section 4 provides descriptive statistics on our dataset as well as findings regarding how privacy policies have been analyzed, what laws and regulations have been investigated, and in which sectors. Section 5 discusses open issues and opportunities for future work. Section 6 concludes the article.

2 Background and Related Work

In this section, we provide an overview of privacy policies and data protection laws and regulations. We then discuss the existing literature reviews on privacy policies.

2.1 Privacy Policies

The notion of a privacy policy emerged as a response to the increasing digitization of personal information and the need to establish guidelines for its handling. It gained prominence with the rise of the internet and e-commerce in the late 20th century. A privacy policy serves as a means to achieve the OECD/FIPPS principles of transparency and openness [2]. Transparency helps individuals understand how their data is handled and enables them to make informed decisions about whether to provide their information. It also fosters trust between organizations and individuals. Openness allows individuals to understand and anticipate how their personal information will be used, which is crucial for maintaining trust and ensuring that data handling practices are aligned with users' expectations.

The shift of privacy policies from self-regulated efforts to mandatory requirements depicts the growing recognition of protecting individual's privacy and personal data [25]. Multiple factors

have contributed to this evolution, including increasing concerns about privacy, global regulatory frameworks such as the GDPR, legislative actions such as CALOPPA, consumer awareness, globalization, and cross-border data sharing.

A privacy policy is a legal document that outlines how an organization collects, uses, discloses, and manages the personal information of individuals and customers. It also describes how the protection of this data is ensured and what controls the users have to limit access to this data. Examples of personal information include name, date of birth, email address, billing and shipping address, phone number, IP address, and so on. Therefore, any organization that collects personal data from their users must have a privacy policy, which should be accessible to the users as a link to a document (typically in the footer section) on the organization's website [1, 3] or their mobile application.

2.1.1 Requirements and Format. The specific requirements for privacy policies can vary, depending on factors such as the jurisdiction, industry, and specific practices of the organization. Similarly, there is no standard structure for a privacy policy document's content. However, it must be organized such that it complies with the principles of the applicable sector or region-specific data protection regulation. Figure 1 shows a snapshot of Google's privacy policy taken from Google's website, which complies with the GDPR. As seen in this figure, below are some key sections commonly included in a privacy policy. However, the format and size of a privacy policy can vary, based on the services provided by an organization.

- **Introduction:** Explains the purpose and scope of the policy and may also include information about the organization's commitment to protecting the privacy and confidentiality of personal information.
- **Types of Information Collected:** Specifies the types of personal information collected by the organization. This may include name, contact information, demographic data, payment details, and browsing history.
- **Methods of Collection:** Describes how the organization collects personal information, whether through website forms, mobile apps, customer service interactions, or other means.
- **Purpose of Collection:** Explains the purposes for which personal information is collected. This could include providing services, processing orders, personalizing user experiences, marketing, and compliance with legal obligations.
- **Use of Information:** Details how the organization uses personal information, including internal purposes, communication with users, improving products or services, and fulfilling legal or regulatory requirements.
- **Disclosure to Third Parties:** Specifies whether and under what circumstances personal information is shared with third parties, such as service providers, business partners, or government authorities.
- **Data Security Measures:** Outlines the security measures implemented to protect personal information from unauthorized access, disclosure, alteration, or destruction. This may include encryption, access controls, regular security audits, and employee training.
- **Data Retention Period:** Indicates how long personal information is retained by the organization and the criteria used to determine retention periods. It also explains the procedures for securely deleting or anonymizing data when it is no longer needed.
- **User Rights:** Informs the users about their rights regarding their personal information and how the users can exercise them. These include the right to access, correct, delete, or restrict the processing of their data.
- **Updates to the Privacy Policy:** Describes the organization's right to update or modify the privacy policy as needed and provides a date for the last revision. Users should be notified of significant changes and their consent should be obtained if required by applicable laws.

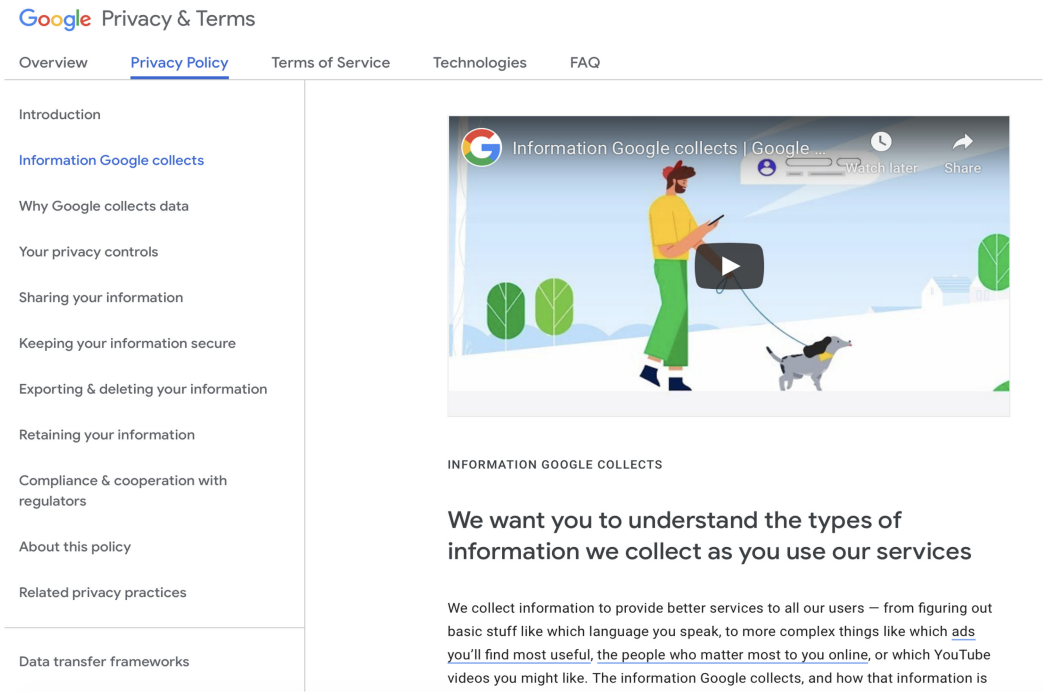


Fig. 1. Snapshot of Google's privacy policy.

— **Contact Information:** Provides contact details for inquiries, requests, or complaints related to the privacy policy or the organization's handling of personal information. This may include email addresses, mailing addresses, or phone numbers.

In terms of the format, a privacy policy is typically presented as a separate webpage on a website or included within a mobile app, accessible via a link in locations such as the website footer, terms and conditions, cookie notice, and account signup [8]. The language should be clear, concise, and written in plain terms understandable to the target audience [3]. Additionally, the policy should be easily navigable, with headings, bullet points, and hyperlinks to facilitate comprehension and access to specific sections [5].

The availability of a privacy policy on the website/application implies that the users consent to the terms defined in it when they visit the website or use the application. Often, the organizations acquire their users' consent by presenting the policy and requesting the users to read and agree to it before they sign up or use their website for the first time. This is done to ensure that customers are fully aware of the data that will be collected, stored, analyzed, managed, or shared by the respective website/application.

2.2 Data Protection Laws and Regulations

Data protection laws and regulations lay down the foundations and define the principles for data privacy and security. These regulations can primarily be divided into three distinct categories, namely, the General Frameworks, Country/ Region-specific Laws and Regulations, and Sector-specific Laws and Regulations. An organization's privacy policy might need to comply with several of these data protection laws.

2.2.1 General Frameworks. Many of the privacy laws originated from the **Organisation for Economic Co-operation and Development (OECD)** privacy guidelines, which were introduced

in 1980 [95]. The OECD framework lays down recommendations for multi-national organizations by governments to hold a responsible business conduct. There are eight privacy principles in the OECD framework [2]:

(1) Collection Limitation: Data collection should occur only with the knowledge and consent of the concerned individual (data subject).
(2) Data Quality: One should only collect information that is relevant and accurate for a particular purpose.
(3) Purpose Specification: The intended use for a particular piece of information must be known at the time of collection.
(4) Use Limitation: Collected data must not be used for purposes other than the ones specified at the time of collection.
(5) Security Safeguards: Reasonable measures must be taken to protect data from unauthorized use, destruction, modification, or disclosure of personal information.
(6) Openness: Individuals should be able to avail themselves of data collection and be able to contact the entity collecting this information.
(7) Individual Participation: The concerned individual should know if their information has been collected and must be able to access it if such data exists.
(8) Accountability: Data collector should be held accountable for failing to abide by any of the above rules.

The OECD principles were later updated, with the most recent version issued in 2013, taking into account recent lessons from non-OECD countries so they continue to maintain their global relevance.

Another general-purpose framework based on the OECD privacy principles that was introduced in 2008 is the **Fair Information Practice Principles (FIPPs)** that were outlined by the United States Federal Trade Commission [53]. FIPPS are guidelines that represent widely accepted concepts concerning fair information practice in an electronic marketplace. However, they are only recommendations for maintaining privacy-friendly, consumer-oriented data collection practices and are not enforceable by law. The FIPPS focuses on the principles of notice/awareness, choice/consent, access/participation, integrity/security, and enforcement/redress.

2.2.2 Regional/Country-specific Regulations. Lately, new and more focused privacy legislations have been enacted regionally that fit to the local privacy requirements in a specific region. A total of 162 countries have therefore enacted their data privacy laws, and another 20 or more are having bills or draft laws to do so [87]. The GDPR is one such legislation created by the EU and is based on the OECD privacy principles [3]. GDPR is globally considered as the most comprehensive and the strongest set of privacy and data protection laws. Failure to comply with its guidelines by an organization that is EU-based or that offers goods/services to a user in the EU can result in a sizable fine. For instance, Google was fined 50 million Euros in 2019 because it failed to provide clarity and transparency on how it handled personal data of its French users for targeted advertising [163]. Similarly, the South African **Electronic Communications and Transactions Act (ECTA)** is an example of a country/region-specific framework that is targeted towards the e-commerce industry in the South African region. It regulates the collection, usage, and disclosure of personal information and promotes consumer trust in online transactions. Another example of a country-specific regulation is the **Personal Data Protection Act (PDPA)**, which is the primary law in Singapore regulating how businesses handle Singapore residents' personal data that they collect. Additionally, multiple US states have, in recent years, adopted comprehensive privacy laws. For instance, the **California Consumer Privacy Act (CCPA)** and **California Online Privacy Protection Act (CalOPPA)** aim to enhance privacy rights and

consumer protection for US residents of California. They require the operators of commercial websites that collect **Personally Identifiable Information (PII)** from California's residents to conspicuously post and comply with a privacy policy that meets specific requirements.

2.2.3 Sector-specific Regulations. Additionally, an organization might need to comply with a sector-specific regulation [6]. For instance, one of the main data protection privacy regulations for the healthcare industry is the **Health Insurance Portability and Accountability Act (HIPAA)**, which applies to health insurance companies, hospitals, and so on. This act has been in use since 1996, and it prevents patients' health information from being disclosed without their consent. It establishes privacy and security standards for **protected health information (PHI)**, thus ensuring confidentiality, integrity, and availability of sensitive medical data. Similarly, the **Children's Online Privacy Protection Act (COPPA)** applies to websites/apps for children and gives parents control over what information the websites can collect from their kids.

All of these frameworks and laws differ from each other in terms of their scope and jurisdiction, principles, requirements, and enforcement mechanisms. For example, OECD principles are applicable to OECD countries, whereas FIPPs is applicable within the US. Similarly, HIPPA and COPPA are the US federal laws. GDPR is specific to the countries within the EU, ECTA is applicable to the South-African jurisdiction, whereas PDPA operates within Singapore. Additionally, the basic requirements and principles of these frameworks also vary, depending on their context and objectives. In terms of enforcement strictness, GDPR is one of the most stringent data protection regulations globally due to its comprehensive scope and substantial fines for non-compliance. Moreover, HIPAA also imposes very rigid requirements for protecting healthcare data with strict privacy and security standards tailored to the healthcare industry.

2.3 Literature Reviews on Privacy Policies

Although several literature reviews have been conducted on privacy policies, they have focused on the depth of a specific aspect rather than the breadth. For instance, Kumaraguru et al. [107] summarized the literature available on privacy policy languages, providing an account of the features, characteristics, and requirements of the languages and describing a comprehensive framework for analysis. This review aimed at enabling organizations to select a privacy policy language from the existing languages and aid them in designing a new language in the future. Moreover, this work was done in 2007, and we add to this literature by analyzing papers after this publication, i.e., after 2007.

Barth et al. [34] performed a literature review on the existing approaches and proposed visualization tools for presenting users' collected information in privacy policies. The goal of this review was to study 15 privacy attributes and rank them according to their perceived importance by users and privacy experts and to compare and contrast their views. We cover visualization and summarization tools in our work, in addition to other aspects such as readability and accessibility analysis.

Del Alamo et al. [64] performed a literature review to specifically look at the existing work done on automated analysis of privacy policies. This review focused on finding out what information can be automatically extracted from privacy policies presented as textual documents and which analysis techniques have been used in this regard. We cover machine learning and natural language processing approaches in addition to manual approaches used for covering other aspects such as user perception, behavior, and comprehension of privacy policies.

Wagner et al. [186] collected a large-scale corpus of privacy policies from 1996 to 2021 and analyzed their length and readability as well as their content in terms of the data practices by using machine learning and natural language processing. This study particularly focused on the changes in response to the recent privacy regulations such as GDPR and CCPA. However, we differ from this work by performing a systematization of existing knowledge on privacy policies

in terms of the various analysis approaches used, the investigated sectors, and the data protection laws/regulations.

Mitchell et al. [126] reviewed 33 articles in the period 2006–2021, to explore privacy concerns in **online social networks (OSN)**, users' behavior associated with privacy preservation strategies and OSN privacy policies. This review aimed to present extensive research on this topic and to highlight any potential gaps. We differ from this study by including articles from several sectors.

3 Methodology

3.1 Research Objectives

The goal of this article is to summarize and extract knowledge from a large corpus of existing research articles on privacy policies. Therefore, we conduct a systematic review of papers published in top peer-reviewed conferences and journals till 31st December, 2023. Our analysis focuses on empirical studies on privacy policies to answer the following research questions:

(1) How have privacy policy texts been analyzed regarding the following aspects?
   — Readability and accessibility of the policy statement;
   — Summarization, visualization tools, and frameworks to help users understand the policy text;
   — Compliance/completeness of policy text with respect to data protection laws/principles;
   — (Mis)Match between organizations' data practices and their privacy policy text;
   — User studies on perceptions, behavior, and comprehension of privacy policies;
   — Use of machine learning and natural language processing in analyzing policy text.
(2) What data protection principles and privacy laws have been investigated, e.g., GDPR and HIPAA?
(3) Which sectors have been investigated, e.g., finance, e-commerce, health, and so on?

3.2 Data Extraction

Our first step was to identify the relevant articles to be analyzed. For this task, we employed the **Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)** framework [128].

3.2.1 Search Strategy. We leveraged four popular bibliographic databases, namely, **Web of Science (WOS)**, Scopus, IEEE Explore, and ACM Digital Library [54]. To ensure that our search is comprehensive, we first identified the relevant keywords and terms through background reading on privacy policy–focused articles and extracted the keywords provided in their title and abstract sections. We also analyzed the keyword variants and synonyms.

After identifying the relevant keywords for our topic, the following search string was designed to search each database individually based on the document titles:

(privacy) AND (policy* OR statement* OR notice OR compliance) AND (study OR analysis OR read* OR eval* OR assess* OR summar* OR present* OR "machine learning" OR "deep learning" OR "natural" OR "nlp" OR customer OR user OR online)

We used the following exclusion criteria in our search:

(1) Articles that are not peer reviewed;
(2) Articles other than conference and journal papers, e.g., posters, workshop papers, reviews, book chapters;
(3) Articles published in a language other than English;
(4) Articles published after 2023.

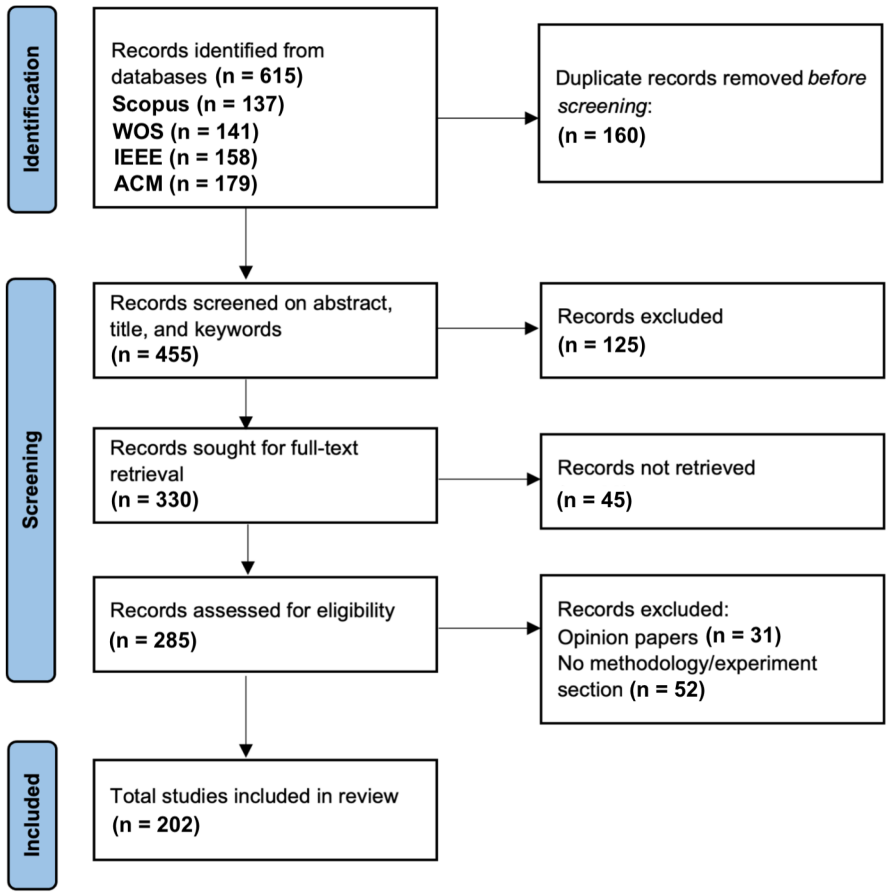


Fig. 2. Article extraction using the PRISMA framework.

Our initial search resulted in 615 articles. The result records from each database were saved as a CSV file. The four CSV files were then combined into one excel sheet, and duplicate entries were removed. This resulted in 455 articles.

3.2.2 Screening and Relevance. An initial screening was performed on the abstract, title, and keywords. Any papers that mentioned privacy policy but whose main focus was on a different topic, such as differential privacy and privacy settings, were removed from the list. This resulted in 330 articles that were then sought for full-text retrieval. We excluded articles for which we were unable to fetch a full-text, either because it was not open-access or we were unable to get a copy of the article from the authors. This resulted in 285 articles. After skimming the full-text articles, posters, short/opinion papers, thesis, or those without a methodology/experiment section were removed. The final list included 202 peer-reviewed articles containing 65 journal articles and 137 conference papers. The complete dataset is available online.[1] Figure 2 shows the various steps in our data extraction process.

3.2.3 Limitations. Several limitations should be considered when interpreting the findings of this review. First, it only includes articles published up to 31st December, 2023, which means that

[1]Paper Dataset Link: https://docs.google.com/spreadsheets/d/1ieLxJWkJBx48G-k61ByiNvctPZ4pLGfiRu3A3v5PqRQ/

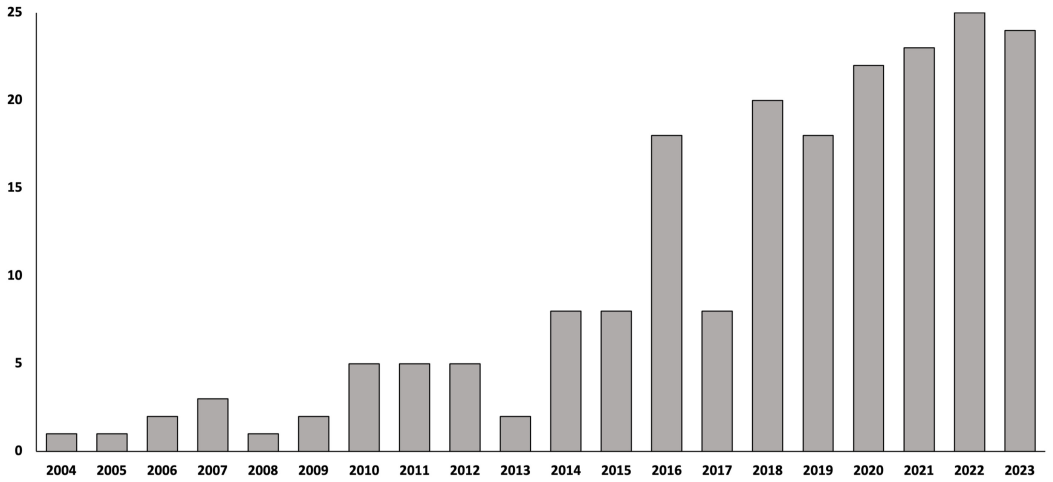


Fig. 3. Distribution of journal and conference papers in our dataset.

a few more recent research papers were not included. Additionally, the review focuses solely on what techniques have been used to analyze privacy policy texts, as well as the investigated sectors and data protection laws and regulations. Based on these research questions, we identified the keywords and search terms to search the relevant literature. However, due to our choice of keywords, there is a risk of omitting relevant literature. Similarly, the initial keyword search was only done on titles and did not include abstracts. These may have led to an undersampling of relevant publications. Finally, the review is limited by the quality and scope of the articles included, as some articles may be more comprehensive or relevant but were not included because of the unavailability of their full-text. Despite these limitations, this review provides a valuable overview of the current state of research in this area and highlights gaps and areas for future research.

4 Results

In this section, we first provide an overview of our dataset including descriptive statistics about the distribution of the analyzed papers across publication venues, geographic locations, and publication years as well as the topics investigated. We then provide details on each analysis method used to investigate privacy policies as well as the investigated data protection laws/regulations, and sectors.

4.1 Dataset Description

Our dataset includes 202 papers focused on privacy policies. A larger percentage of these articles were published at peer-reviewed conferences (67.82%, N=137) compared to peer-reviewed journals (32.17%, N=65). Figure 3 shows the distribution of these papers across publication years. A majority of the papers in our dataset were published between 2016 and 2023, which signifies the continued importance of this topic. This increased considerably after the GDPR went into effect in 2018. A larger percentage of the privacy policy research was carried out by researchers in 38 countries, with United States (41.1%) leading, followed by Germany (9.4%), China (7.9%), Canada (7.9%), and United Kingdom (5.4%) (see Figure 4).

The most investigated topics were the use of machine learning, deep learning, and NLP to analyze privacy statements (42.57%, N=86) as well as to investigate their completeness and compliance with privacy laws and regulations (35.6%, N=72), whereas the least investigated topic was exploring whether there is a mismatch between an organization's privacy policy statement and its actual

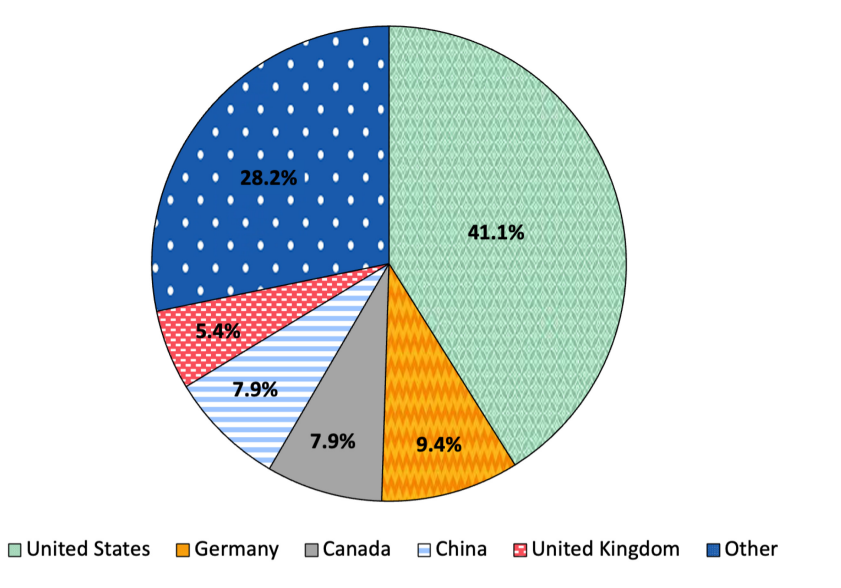


Fig. 4. Distribution of countries where privacy policy research is being conducted.

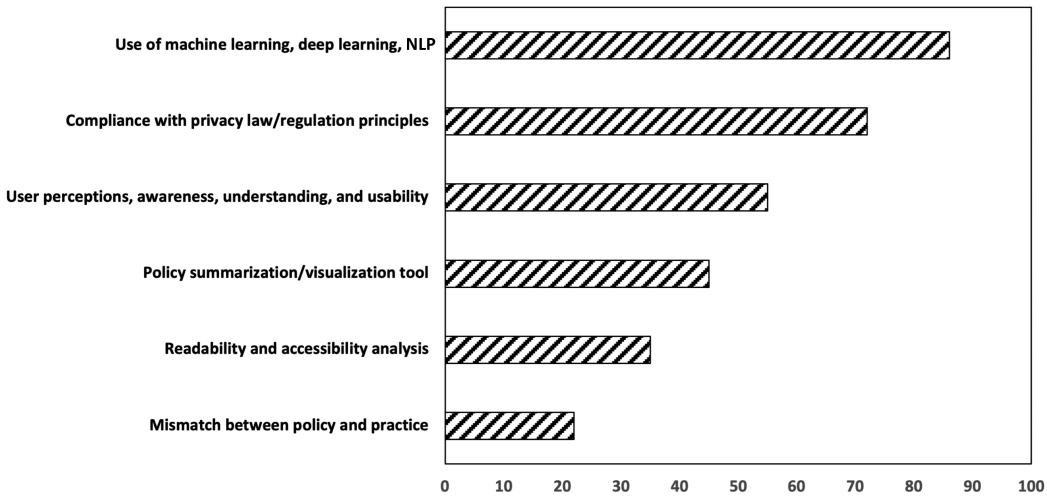


Fig. 5. Distribution of investigated topics in our dataset.

privacy practices (10.90%, N=22). Figure 5 shows a distribution of the investigated topics in our paper dataset.

4.2 RQ1: Analysis Methods

4.2.1 Readability and Accessibility of Policy Text. The readability of a privacy policy determines whether its content is written in a clear and concise language (i.e., with limited technical jargon). Typically, a score is computed to estimate the educational level required for users to understand what personal information is being collected from them and how this information is being used,

stored, and shared [200]. A good readability score indicates that the privacy policy text is easy to understand for an average user.

17.33% (N=35) of the papers in our dataset investigated the readability and accessibility of privacy policy text. The methodology used consisted of compiling a dataset of privacy policies and then computing readability scores on one or more metrics using online readability calculation tools, R packages such as koRpus and TreeTagger in R, and Python's **Natural Language Toolkit (NLTK)** etc. [4, 42, 75].

An organization's privacy policy is primarily accessed through a link in the footer section of the website, via the navigation menu, or through the settings section of a mobile app. A majority of the readability-focused articles in our dataset additionally assessed the accessibility of privacy policy, i.e., whether a privacy policy was present on the website/mobile app and how easy is it for the users to locate it.

Readability Metrics. There are various readability metrics that can be used to measure the readability of a piece of text. Some of the most common measures include the Flesch–Kincaid Grade Level, the Flesch Reading Ease, and the Gunning Fog Index [200].

- Flesch–Kincaid Grade Level is based on the average number of words and syllables per word. This metric rates a text on the US school grade level. The values vary from 0 to 18, where 18 represents the most difficult text. A value of 8.0 is recommended for the English content to be easily understood. It means that an eighth-grader can easily understand the document.
- Flesch Reading Ease was developed by Rudolf Flesch in the 1940s, and it scores a text on a scale of 0 to 100, with 100 being the highest readability score. A score between 60 to 80 is equivalent to school grade level 8, which means that the text should be fairly easy for an average adult to read.
- Gunning Fog Index is a test that measures the readability of a text by counting the number of words and sentences in a passage. It then uses a formula to calculate the average number of words per sentence. The higher the number, the more difficult the text is to read. A Gunning Fox Index of 7–8 is ideal, and anything higher than 12 is too complex for most people to read.

A majority of the readability-focused papers in our dataset leveraged a combination of existing metrics such as **Flesch-Kincaid Grade Level (FKGL)**, **Flesch Reading Ease Score (FRES)**, **Gunning-Fog Index (GFI)**, **Simplified Measure of Gobbledygook (SMOG)** Index, **Automated Readability Index (ARI)**, and word count to analyze the readability of policy texts. Four papers mentioned using **Coleman-Liau Index (CLI)**, and **New Dale-Chall Score (NDC)** but did not provide any scores. Two of the papers created their custom criteria for evaluation, and five of the papers did not provide any details on the criteria used to measure readability.

Table 1 lists each readability metric, its score, and the type of policies evaluated. Analysis across various sectors shows that even after approximately two decades, the readability of privacy policies remains low. Therefore, the content is still too difficult for the target reader to understand. For instance, the FKGL for banking websites, online social networks, telecom companies, and other high-traffic websites is greater than 14, which means that they require more than 14 years of education to understand the policies [60, 94, 102, 164]. Similarly, the FRES score across sectors is extremely lower (20–50) than the target value of 70 [20, 164]. Other readability metrics such as GFI and ARI have values (10–16) significantly deviating from the expected values of 7–8 [20, 59]. Overall, healthcare mobile app policies seem to have relatively good readability scores compared to the policies of other sectors with FKGL close to the expected value of 8 and SMOG index close to the expected value of 5–8 [77, 204].

In terms of the word count, the policies seem to have increased in length overtime, especially post GDPR and CCPA enforcement [186]. For instance, the average word count of healthcare

Table 1. Description of Readability Focused Papers in Our Dataset

Readability Metric (Expected Score)	Average Score	Type of Policies	Reference
Flesch-Kincaid Grade Level (8)	10.8, 12.1	50 banks, 54 mobile money services in US	Bowers et al. 2017 [44]
	12	5 digital content vendors of USA libraries	Lambert et al. 2015 [111]
	14	284 South Asian websites in 10 sectors	Javed et al. 2020 [93]
	8.5	27 free mobile health apps	Farooq et al. 2020 [77]
	16	11 Croatian news websites	Alic et al. 2020 [20]
	14.05	35 online health social networks	Savla et al. 2012 [164]
	9.37	7 COVID contact-tracing apps	Zhang et al. 2020 [204]
	13.8	15 menstruation-tracking iOS apps	Fowler et al. 2020 [79]
	13.7, 13.6	319 iOS & 69 Android mental health apps	Robillard et al. 2019 [158]
	12.78	64 Android & iOS apps for minors	Das et al. 2018 [63]
	12.37	Google, LinkedIn, Twitter, & Facebook	Winkler et al. 2016 [189]
	14.60	101 location/mobility service companies	Cottrill et al. 2014 [60]
	14.05	70 5G networks in UK	Zhang et al. 2023 [203]
	14.2,14.0	64 High traffic, healthcare websites	Jensen et al. 2004 [94]
	14.68	14 South African banks	Keshav et al. 2010 [102]
Flesch Reading Ease Score (70)	33.58	5 digital content vendors of US libraries	Lambert et al. 2015 [111]
	50.25	27 free mobile health apps	Farooq et al. 2020 [77]
	30.87	11 Croatian news websites	Alic et al. 2020 [20]
	20.25	35 online health social networks	Savla et al. 2012 [164]
	50.71	7 COVID contact-tracing apps	Zhang et al. 2020 [204]
	38.95	15 social networks for kidney disease	Costello et al. 2015 [59]
	36.86,43	Top 100 Fortune Global companies	Boldt et al. 2019 [42]
	32.64	14 South African banks	Keshav et al. 2010 [102]
	50	2,500 popular Android apps	Yu et al. 2018 [196]
	39.6	6 companies	McDonald et al. 2009 [122]
Gunning-Fog Index (7–8)	10.64	27 free mobile health apps	Farooq et al. 2020 [77]
	10.64	7 COVID contact-tracing apps	Zhang et al. 2020 [204]
	14.7	15 social networks for kidney disease	Costello et al. 2015 [59]
Simplified Measure of Gobbledygook (5–8)	9	27 free mobile health apps	Farooq et al. 2020 [77]
	15.54	11 Croatian news websites	Alic et al. 2020 [20]
	8.8	7 COVID contact-tracing apps	Zhang et al. 2020 [204]
Automated Readability Index (8)	16.74	11 Croatian news websites	Alic et al. 2020 [20]
	14.29	70 5G networks in UK	Zhang et al. 2023 [203]
Lix Readability (30–35)	68.6	11 Croatian news websites	Alic et al. 2020 [20]
Dave-Chall (6–6.9)	8.7, 12.9	Top 100 Fortune Global companies	Boldt et al. 2019 [42]
Cloze Test (.60)	NA	5 big Internet services	Ermakova et al. 2014 [71]
	0.39	10 popular websites	Singh et al. 2011 [167, 168]
Total Word Count	1492, 1374	50 banks, 54 mobile money services in US	Bowers et al. 2017 [44]
	1318	5 digital content vendors of US libraries	Lambert et al. 2015 [111]
	3057	27 free mobile health apps	Farooq et al. 2020 [77]
	2031	7 COVID contact-tracing apps	Zhang et al. 2020 [204]
	2425	64 Android & iOS apps for minors	Das et al. 2018 [63]
	1849, 2661	325 Pre-, post-GDPR Android mobile apps	Anikeev et al. 2021 [27]
	2220, 1540	Top 100 Fortune Global companies	Boldt et al. 2019 [42]
	611.44	14 South African banks	Keshav et al. 2010 [102]
Complex Word Count	17.91	7 COVID contact-tracing apps	Zhang et al. 2020 [204]
No. of Words in Sentence	24.3	15 menstruation-tracking iOS apps	Fowler et al. 2020 [79]
Custom Criteria	NA	115 cloud vendors	Kotal et al. 2020 [104]
	NA	Top 100 New Zealand organizations	Tjhin et al. 2016 [182]

mobile app policies in 2020 is 3,057, whereas the average word count of banking websites policies in 2010 is 611.

Automated Analysis. Two papers proposed completely automated approaches for readability analysis of policies on a large scale [66, 75]. The first step is web scraping and crawling to extract policies from the top/popular websites' HTML by downloading the HTML, extracting potential privacy policy URL, and then extracting text from that URL. Next, machine learning classifiers such as Naïve Bayes, **Decision Tree (DT)**, and **Maximum Entropy (ME)** [42] are used to classify a document as privacy policy or not and then correspondingly store it in a database. Finally, tools such as Python's NLTK [4, 42, 75] are used to calculate various readability measures on the whole policy or separately for each section. For the latter, word-embedding models such as Word2vec and GLoVe are used to extract word vectors for each section [196]. These word vectors are then used with a multiple attribute-level CNN-based classifier (ATT-BI-GRU) [66] to classify each privacy policy document into 12 sections. Readability metrics are then calculated on each section.

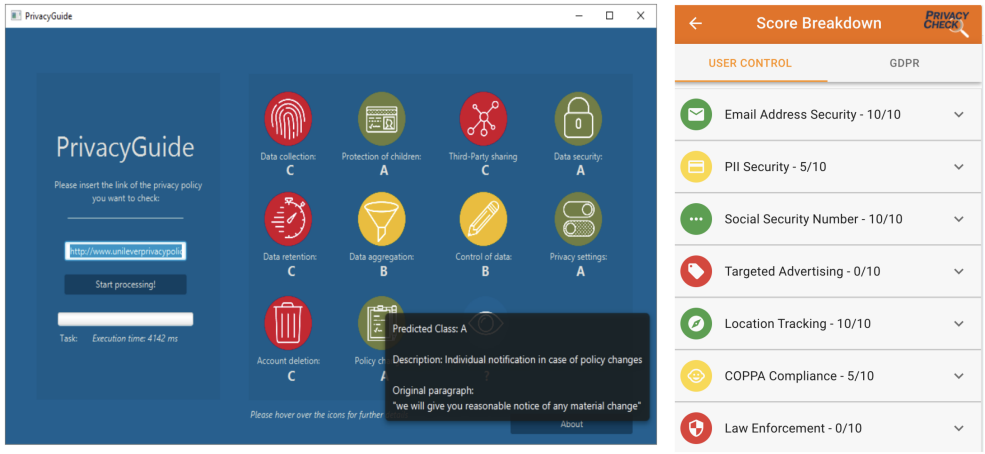
Accessibility Metrics. The papers focused on assessing readability or compliance/completeness of privacy policies have also investigated their accessibility using a simple metric, i.e., whether the organization's website/mobile app provides a privacy policy or not [102]. Others used a custom scoring criteria based on how easy it is to find it on the website, i.e., within how many clicks [93, 164].

Our analysis shows that mobile applications have lower accessibility of privacy policies as compared to websites. The types of apps studied include mobile money, fitness, menstruation tracking, and mental health mobile apps [44, 79, 137, 158, 160]. The investigated websites include healthcare, online social networks, and popular/high traffic websites in other sectors [59, 93, 164].

4.2.2 Summarization and Visualization Tools to Help Understand Policy Text. Several studies have shown that users do not read privacy policies, since they are lengthy, contain technical jargon, and have an average reading time of about 12 minutes [121, 135]. Therefore, numerous efforts have been made towards helping users understand the content of these policies via tools that summarize key information in the policies in the form of text and visualizations. 22.27% (N=45) of the papers in our dataset proposed (1) a policy content summarization tool that provides a concise summary of data collection and sharing (N=15); (2) a visualization tool that additionally uses policy completeness scores, animations, icons, colors, mappings, and so on, to convey this information (N=15); (3) an interactive Querying tool to answer users' questions and assist them in understanding the content of a privacy policy (N=4); or (4) a framework that can be used towards this goal by helping annotate the policies, author better policies, extract ontologies, or convert them into a machine readable format (N=11).

Table 2 lists the tools and frameworks proposed in our investigated papers. Various policy visualization tools have been proposed. For instance, Privacy Guide comprises indicators representing each privacy policy section such as "Data collection" and the associated risk level (via colors) [180] (see Figure 6(a)). Some systems visually summarize the completeness score of a privacy policy against a specific regulation's principles using a web browser extension. For instance, the completeness analyzer and Privacy Check show a completeness score and use icons to show whether a data protection principle such as "Data Retention" is covered by the privacy policy [58, 199] (see Figure 6(b)). Other proposed presentation formats include video, nutrition label, and expandable grid.

Machine learning and NLP were used in four of the visualization tools mainly for annotation purposes. The state-of-the-art approach for annotation consists of using a dataset such as the **OPP-115 Corpus (Online Privacy Policies, set of 115)** [188] to train machine learning



((a)) Privacy Guide ((b)) Privacy Check

Fig. 6. Snapshots of visualization tools with scores.

classifiers such as Naive Bayes and Random Forest to divide the individual sections of a privacy policy into categories [49].

Among the summarization tools, some focused on a specific aspect such as summarizing **personally identifiable information (PII)** [199] or a specific section of the policy such as "Data Collection" [97]. Others focused on annotating/extracting important sentences using keyword matching–based sentence classification, categorizing the policy into 9–10 categories, or app source code into related or predefined categories and labels [15, 81, 92, 99, 183]. Thirteen of these summarization techniques used data mining, machine learning, and NLP.

Four papers proposed structured and freeform querying tools to allow users to interact with and extract specific information from the policy in real time [90, 138]. All of these tools leveraged NLP.

Others proposed frameworks and tools to annotate, extract ontologies, author, and convert privacy policy into a machine-readable format (XML) [47, 162] or formal representations that can be utilized to help users understand policies or help them if organizations improperly disclose their privacy-sensitive behaviors [26]. Three of these approaches used NLP and machine learning.

Automated Analysis. The state-of-the-art in automated policy summarization includes several steps [15, 90]. The first step is web scraping and crawling to extract policies from the top/popular websites' HTML by downloading HTML, extracting potential privacy policy URL, and then extracting text from that URL. The next step is pre-processing to remove the stop-words, word lemmatization, and stemming. Then, paragraph/segment representation is done using various text representation techniques such as word mapping, count vectorizer, TFIDF, Doc2Vec, GLoVe, **Universal Sentence Encoder (USE)**, and WordPiece to prevent manual annotation [31]. Finally, multi-label classification is performed using machine and deep learning algorithms to detect paragraph-level categories. These algorithms include **Logistic Regression (LR)** [69], **Support Vector Machine (SVM)**, **Random Forest (RF)** [180], **Convolutional Neural Networks (CNN)** [103], **Deep Neural Network (DNN)** [90], and **Bidirectional Encoder Representations from Transformers (BERT)** [69].

4.2.3 Policy Text Compliance/completeness with Respect to Data Protection Principles. A majority of the papers in our dataset, 35.6% (N=72) investigated whether a privacy policy is complete

Table 2. Tools and Frameworks Proposed by Researchers to Help Understand Policy Text

Tool/Framework Name	Type	ML/NLP	Reference
Privacy Guide	Visualization	Yes	Tesfay et al. 2018 [179, 180], Bracamonte et al. 2019 [45, 46]
PrivacyCheck	Visualization	Yes	Zaeem et al. 2018 [199]
Policy Completeness Analyzer	Visualization	Yes	Costante et al. 2012 [58], Guntamukkala et al. 2015 [89]
Web Extension–based Policy Visualization	Visualization	Yes	Brunotte et al. 2022 [49]
PPV Model	Visualization	No	Ghazinour et al. 2016 [85]
Video and Usable Policy Format	Visualization	No	Faurie et al. 2020 [78]
Document Card and Word Bridge	Visualization	No	Soumelidou et al. 2020 [171]
Visual Interactive Policy Using Nutrition Label Format	Visualization	No	Reinhardt et al. 2021 [156], Kelley et al. 2010 [101]
3 standardized privacy policy formats	Visualization	No	McDonald et al. 2009 [122]
P3P policy presentation tool using Expandable Grid	Visualization	No	Reeder et al. 2008 [155]
Automated Policy Summarizer	Summarization	Yes	Tomuro et al. 2016 [183]
Domain-aware Extractive Model	Summarization	Yes	Keymanesh et al. 2020 [103]
Text Extraction on “Data Collection”	Summarization	Yes	Joshi et al. 2016 [97], Alohaly et al. 2016 [21]
Domain Ontology–based Text Extraction	Summarization	Yes	Kaur et al. 2022 [99]
NLP-based Summarizer	Summarization	Yes	Perera et al. 2021 [145]
BERT extractive summarizer	Summarization	Yes	Elluri et al. 2021 [69]
Privacy Captions for Android App Source Code	Summarization	Yes	Jain et al. 2022 [92]
“Human-in-the-Loop” Approach to Policy Annotation	Summarization	Yes	Gebauer et al. 2023 [83]
PriVot	Summarization	Yes	Atapattu et al. 2021 [31]
TLDR (Automated Annotation)	Summarization	Yes	Alabduljabbar et al. 2021 [15, 16]
Classification of sentences into labels	Summarization	Yes	Fukushima et al. 2018 [81]
Contextual Privacy Policies	Summarization	No	Ortloff et al. 2020 [141]
Ppmark	Summarization	No	Pontes et al. 2017 [149]
PriBot	Querying Tool	Yes	Harkous et al. 2018 [90]
PrivOnto	Querying Tool	Yes	Oltramari et al. 2018 [138]
Machine Reading Comprehension & Open Domain Q/A	Querying Tool	Yes	Sanghavi et al. 2022 [162]
Privacy Policy Question answering model	Querying Tool	Yes	Thotawaththa et al. 2021 [181]
RSL-IL4Privacy	Framework	Yes	Da Silva 2016 [62]
Automatic Ontology Extraction via Hyponymy	Framework	Yes	Evans et al. 2017 [73]
Policy conversion to standardized language (S4P)	Framework	Yes	El et al. 2023 [68]
Privacy Policy Specification	Framework	No	Mohammadi et al. 2019 [86]
IoT Device Privacy Policy Understanding	Framework	No	Onu et al. 2020 [139]
Formal Modeling	Framework	No	Manna et al. 2018 [118]
Privacy Policy Language PILOT	Framework	No	Pardo et al. 2019 [143]
Policheck	Framework	No	Andow et al. 2020 [26]
SPARCLE system (Policy parsing & conversion to XML)	Framework	No	Brodie et al. 2006 [47]
Privacy Policy Rule Authoring System	Framework	No	Karat et al. 2006 [98]

with regards to the applicable data protection law/regulation principles, i.e., how well does it comply with the principles of a specific data protection law/regulation.

A common approach used to evaluate the completeness of a policy text comprised of compiling a dataset of privacy policies of websites/apps from a specific sector(s) [93]. Next, an applicable data protection law/regulation for that sector/country/region was selected and a scoring criteria was formulated based on the principles in the corresponding law/regulation (GDPR being the most common (N=33)). Section 4.3 discusses the data protection laws/regulations and frameworks investigated in our dataset. Finally, a score was computed for each privacy policy in the dataset. For instance, Javed et al. computed a score out of 11 based on the 11 privacy principles of the GDPR [93]. Eight papers performed content analysis on the policies by designing a custom criteria on privacy/security principles instead of a specific law/regulation [10, 32, 59, 100, 125, 148, 158, 160] .

Automated Analysis. A subset of these papers (N=30) used **Natural Language Processing (NLP)** or supervised machine learning and deep learning approaches to automate the completeness evaluation process by annotating/extracting features/labels and computing a compliance score with high accuracy. The privacy policies were first manually annotated for preparing the training dataset. The sentences/paragraphs in the policy were labeled based on which privacy principles they belong to. For instance, for GDPR, the sentences were classified into 10 labels corresponding to the 10 privacy principles. Then, (1) NLP classifier such as BERT-based model is trained and used to score the compliance of a privacy policy document by checking whether the 10 rules/labels appear or not [203]; (2) feature vectors such as TD-IDF vectorizer, word embeddings are used together with machine learning classifiers such as XGBoost, Random Forest, Extra Trees, Logistic Regression, KNN, Decision Tree, and SVC to compute a compliance score [178, 184].

Other approaches did not manually annotate policies. Akanfe et al. [13] used text mining and NLP techniques such as **term frequency-inverse document frequency (tf-idf)** and **Latent Dirichlet Allocation (LDA)** to extract semi-structured information components from the privacy policy text. Next, text categorization and topic modeling was done to create a compliance index based on the 10 dimensions of GDPR. Mori et al. used word embeddings to convert the policy content into vectors to train and build subsequent CNN classifiers, which classified policy vectors into various categories (employed from the OPP-115 dataset introduced by Wilson et al. [188]) or created based on the applicable privacy principles) [10, 129].

Appendix A provides details on privacy compliance/completeness-focused papers in our dataset. The types of websites/mobile apps for which the privacy policy completeness was analyzed include personal health record platforms [51, 59, 119], health and fitness apps (mental health, cancer, pregnancy monitoring, dementia, female health) [33, 39, 56, 70, 158, 160, 164, 195, 201], banking/funding, and mobile money services [13, 44, 102, 165, 184], carmakers' apps [41], digital content providers of libraries [111, 117], e-government websites [30, 134], Internet service providers and 5G networks [80, 203], online social networks and crowdsourcing platforms [65, 124, 125], e-commerce websites [69, 114], IoT devices/wearable apps and manufacturers [32, 67, 110, 153], cloud computing/storage applications and location services [60, 82], virtual reality apps [202], virtual assistant apps [193], companies [9, 24, 129, 146, 157, 205], third-party libraries in apps [207], Alexa popular/free content websites from multiple sectors [10, 17, 36, 55, 148, 151, 154, 186], top/most downloaded mobile apps from multiple sectors [55, 116, 120, 178].

Our analysis showed that in some cases such as the top Fortune 100 companies had high compliance with EU-US Safe Harbour Framework [42]. Similarly, large public companies have better compliance with GDPR and FTC FIPPS than small companies [9, 146]. However, Malaysian companies in the private sector have better compliance with Personal Data Protection Act than companies in the government sector [57]. Similarly, policies of websites post GDPR enforcement have better compliance than before GDPR [36]. The remaining sectors had relatively low completeness.

These include mobile money services and banks, crowdsourcing platforms, e-government websites, Internet service providers, and healthcare platforms/social networks apps.

Highlighted issues from the completeness analysis include (1) missing information on children's data protection, data retention and portability, personal data usage and sharing with aggregators; (2) data anonymity is not ensured; (3) more notice is provided than choice; (4) concerns about lack of consent, transparency; and (5) poor information security control.

4.2.4 User Perceptions, Behavior, and Comprehension of Privacy Policies. 27.22% of papers in our dataset (N=55) conducted user studies to investigate user perceptions, behavior, and comprehension of privacy policies. Both quantitative and qualitative methods were used in these papers. Appendix B summarizes the design, goal, demographics, and sample sizes of each user study.

The quantitative methodologies include controlled lab and online experiments (N=26) that used a between-subject or a pre/post design to investigate (1) how users read the policies (e.g., through eye-gaze tracking); (2) how well they comprehend or trust them; or (3) the usability and effectiveness of policy summarization/visualization tools/presentation formats. Several papers (N=20) conducted large-scale online surveys to explore the factors impacting readability and trust in privacy policy, as well as user behavior, and perception by leveraging various theories. **Partial Least Squares Structured Equation Modeling (PLS-SEM)** was also utilized for analysis in several of these online survey-based studies.

The qualitative methods include open-ended surveys (N=2), focus groups (N=1), structured and semi-structured interviews (N=2), to understand user perceptions about different policy presentation formats as well as the factors that impact their motivations to read privacy policies. One paper performed heuristic evaluation on privacy policy presentation tool using the STARP framework.

Our analysis of the user studies related to privacy policy comprehension shows that the content readability and incorporation of visualization/video format/icons such as privacy seals on the policy impacts its importance and user trust in the service provider and creates better awareness about privacy [12, 37, 38, 45, 46, 71, 72, 78, 90, 113, 141, 149, 168, 171, 206]. However, culture does impact the relationship between the content of privacy policy and privacy concern/trust [191]. Certain presentation environments are better for readability, such as desktop compared to mobile [167]. Various factors impact how much time the users will spend reading the policy. These include policy length, users' online privacy literacy, visibility, specificity, and whether it is presented to the users or they have to find and read it [50, 61, 123, 127, 136, 173, 176].

Various theories and models were also leveraged to understand user attitude and behavior towards privacy policies. These include the neutralization theory [23], control/restricted theory, cognitive theory of multimedia [113], deterrence theory [109], Technology Acceptance Model [149], Privacy–Trust–Behavioral Intention Model [191], Heuristic-Systematic Model [194], protection motivation theory, and the theory of reasoned action [166].

In terms of the investigated populations, 19 user studies focused on undergraduate and graduate university students or participants from crowdsourcing websites such as MTurk. Other studies recruited nurses and medical interns, seniors, IT staff, school and bank employees, or online users. These participants were recruited from various countries, including Germany, Greece, Ireland, Netherlands, Russia, Brazil, Canada, US, Japan, Taiwan, South Africa, Israel, and Saudia Arabia. Therefore, mostly, the participants were from the West (Europe, US, Canada) and East Asia. Other countries, such as those in the Middle East, South Asia, and Africa, were relatively understudied.

4.2.5 (Mis)Match between Organizations' Data Practices and Their Privacy Policy Text. Even if an organization's privacy policy is complete with regards to the principles in the relevant data protection law/regulation, it does not guarantee that their actual data collection and processing practices match with their statements. Therefore, this topic is important, but relatively harder to

investigate, as it requires one to first gather actual data being collected by the organization from its customers and then compare it with the information mentioned in their policy text. Only 10.90% papers (N=22) in our dataset investigated this topic. Nearly all studies showed some mismatch between policy statement and practice. Table 3 shows details of papers focused on mismatch between privacy policy and practice.

A majority of these papers (N=16) focused on detecting privacy policy violations/leaks in the top, popular, or most downloaded Android/iOS mobile applications (including those hosted on cloud and fog computing platforms) to find out how trustworthy their privacy policies are [26, 31, 70, 76, 108, 132, 152, 159, 169, 170, 177, 185, 187, 192, 196, 197, 207]. Their approach consisted of first extracting PII-related information mentioned in the privacy policy documents of apps, either manually or using crowdsourcing websites (such as MTurk) and NLP. Then, the runtime behaviors of mobile applications were analyzed (using network traffic analysis, static, and dynamic analysis tools such as AppAudit and Flowdroid) to find the actual transmitted PII (such as location or device ID). Various similarity techniques were then used to find the match between the mentioned and actual transmitted PII information.

Nwaeze et al. [134] investigated the PII collected by e-government services in five countries of Anglophone West Africa by inspecting their privacy policies. Passive security reconnaissance was then performed (using Wireshark) on these websites to analyze the security of the PII transmission. Their results showed weak protection of PII.

Subahi et al. [174] and Yu et al. [198] investigated the compliance of IoT device's data disclosure with their privacy policy agreement. They studied various IoT manufacturers and platforms and captured the data traffic (via Wireshark) between the IoT device and the cloud, between the IoT device and its mobile application, and between the IoT application and the cloud, and analyzed those packets for various features extracted from the corresponding privacy policies.

Anthonysamy et al. [28] qualitatively analyzed whether the privacy policies are effectively reflected in the privacy controls of social networking websites (such as Facebook, MySpace, Twitter, and PerfSpot).

Joshi et al. [96] developed an automated privacy compliance approach for PII on the cloud by using blockchain to capture each data operation and validate it against the content of privacy policy.

Automated Analysis. The first step in automated violation detection is ontology and mapping development to convert code or policy text to phrases in ontology. Next, privacy policy analysis extracts all declarations related to the user privacy collected by app and third parties and so on. These declarations are converted into three-tuples in the form of (entity, action, data type) [177, 197]. Static or dynamic analysis is then used to inspect the app's APK and detect the data flows where the app shares user private information. Next, tuples in the form of (entity, collects, data type) are constructed [31, 196]. Finally, the two sets of tuples obtained from privacy policy analysis and static analysis, respectively, are compared to detect inconsistency between them.

4.2.6 Use of Machine Learning and NLP in Analyzing Policy Text. A majority of the papers in our dataset, 42.57% (N=86) utilized machine learning, deep learning, and NLP to automate privacy policy analysis. Several papers used automation in one or more aspects, i.e., to accurately label/annotate sections/content of the privacy policy text to build a labeled policy corpus for aiding in summarizing/visualizing the content for end-users, to score its completeness against privacy principles specific to a data protection law/regulation, or to find out violations between policy statement and practice.

For evaluating policy completeness, a training dataset was first created on a set of privacy policies that were labeled based on specific privacy principles such as those in GDPR. Next, various supervised classification algorithms were used to correctly score the privacy policies in

Table 3. Description of Papers Focused on Mismatch between Privacy Policy and Practice

Type of Policies	Approach	ML/NLP	Findings	Reference
6 IoT platforms	Traffic analysis	Yes	23% of the collected sensitive information was not stated in the policy	Yu et al. 2021 [198]
12 Cloud/fog computing apps	Traffic analysis	Yes	Approach identifies apps' encrypted traffic with 86% accuracy	Farhadi et al. 2021 [76]
4 Social media websites	Manual qualitative mapping	No	Principles addressed in privacy policies are not reflected in the privacy controls	Anthonysamy et al. 2011 [28]
11 IoT manufacturers	Network traffic analysis between device & cloud	No	No device fully adheres to their policy	Subahi et al. 2018 [174]
Cloud apps	Automated compliance using Blockchain	Yes	NA	Joshi et al. 2019 [96]
24 E-government services in West Africa	Network traffic analysis	No	Mismatch between policy and data practice	Nwaeze 2017 [134]
120 Popular mobile apps	Static information flow analysis	Yes	Approach detected more than 21 strong violations	Wang et al. 2018 [187]
13,796 Android apps		No	42.4% apps incorrectly disclose their privacy-sensitive data flows	Andow et al. 2020 [26]
30 Android + iOS apps	Manual qualitative analysis	No	Both Android and iOS app data handling practices do not fully comply with their privacy policies	Kununka et al. 2018 [108]
Android apps		Yes	60% apps do not disclose at least one permission-related data collection in privacy policies	Rahman et al. 2022 [152]
40 Android apps	APK and permission analysis	Yes	Approach has high accuracy in detecting conflicts	Wu et al. 2023 [192]
2,500 Popular Android apps	Static analysis	Yes	74% apps have some mismatch	Yu et al. 2021 [196]
15K Android apps	Traffic analysis	Yes	Automated approach has 67.4% accuracy	Atapattu et al. 2021 [31]
9K Random Android apps	Identifying the library triggering data transfer	Yes	81.12% apps fail to transparently declare the recipients of personal data in their privacy policies	Rodriguez et al. 2023 [159]
9K Android apps	Analyze APK files and privacy policy	Yes	Approach has 81% recall	Narksenee et al. 2019 [132]
11 Female health apps	Traffic analysis	No	Data aggregators mentioned in policies were absent in collected data	Erickson et al. 2022 [70]
Most downloaded Android apps	Static analysis and dynamic analysis	Yes	Only 1% apps' data collection match policy description	Verderame et al. 2020 [185]
1,197 Android apps	Static analysis	Yes	23.6% apps have violations	Yu et al. 2016 [197]
247 TPLs, & 641 host apps	Analyze bytecode of TPLs & host apps	Yes	47% TPLs miss disclosing data usage. 65% host apps miss disclosing interactions with TPLs	Zhao et al. 2023 [207]
Top 50 Android apps	Comparison of data collection code and related description in privacy policy	No	NA	Slavin et al. 2016 [169]
477 Android apps	Link policy phrases to API methods	Yes	341 potential privacy policy violations	Slavin et al. 2016 [170]
1,000 popular Android apps	Static analysis	Yes	Apps violate user privacy disclosures to third-parties	Tan et al. 2023 [177]

Table 4. Machine Learning and Natural Language Processing Approaches Used in Our Investigated Papers

Algorithm Category/Name			Use
Machine Learning	Supervised Learning	Decision Tree (DT)	Policy completeness/Readability
		Naive Bayes	Policy completeness/Readability
		Logistic Regression (LR)	Policy completeness
		Maximum Entropy (ME)	Readability
		Support Vector Machines (SVM)	Policy completeness
		Random Forest	Policy completeness
		Ridge Regression (Ridge)	Policy completeness
		k-Nearest Neighbor (k-NN)	Policy completeness
		Gradient/Adaptive Boosting	Policy completeness
		Bayesian Inference	Policy completeness
	Deep Learning	Convolutional Neural Networks (ATT-BI-GRU)	Policy completeness/Readability
		Long Short-Term Memory (LSTM)	Policy completeness
		bidirectional Long Short-Term Memory	Policy Q/A tool
NLP	Semantic analysis using Word Embeddings		Policy annotation/summarization
	Text Mining (tokenization, lemmatization, stemming, etc.)		Policy preprocessing
	Keyword extraction (TF-IDF, RAKE, TextRank, and AlchemyAPI)		Policy annotation/summarization
	Text representation (word mapping, Doc2Vec, USE, WordPiece		Policy annotation/summarization
	Topic Modeling (Latent Dirichlet Allocation (LDA))		Policy annotation/summarization
	Sequence Labeling		Policy annotation/summarization
	Parts of Speech Tagging		Policy annotation/summarization
	Ontology Development		Policy completeness/annotation
	Bag of Words for Text Labeling		Policy completeness/annotation
	BERT Model		Policy completeness/annotation

the test dataset to evaluate their completeness. The investigated supervised machine learning algorithms include Support Vector Machines, k-Nearest Neighbor, Random forest, Naïve Bayes, Decision Tree, Logistic Regression, Neural Networks, Gradient Boosting, and Adaptive Boosting [42, 89, 130, 132, 180, 184, 185, 199]. Similarly, deep learning algorithms such as **Convolutional Neural Networks (CNN)** [118], **Long Short-Term Memory (LSTM)**, and bidirectional Long Short-Term Memory biLSTM) [35] were used.

The NLP approaches were used for information extraction and topic modeling to annotate privacy policies to build a labeled dataset or to use them to summarize policy text for the end-user. The explored NLP approaches include text mining and preprocessing [96], topic modeling using LDA [100], semantic analysis using word embedding and word vector representation [106, 130], keyword extraction [13, 89], sequence labeling [188], word segmentation [198], bag of words for text labeling [52], parts of speech tagging [40], ontology [99], and BERT language model [203].

Table 4 summarizes the various machine learning and natural language processing approaches used in our dataset.

4.3 RQ2: Data Protection Laws and Regulations

The data protection laws and regulations investigated in our dataset mainly belonged to three categories, i.e., Country or Region-specific, Sector-specific, and General Frameworks, and were used to investigate the completeness checking of privacy policies. Table 5 shows a distribution of these data protection laws and regulations.

Among the country/region-specific laws and regulations, the European GDPR was the most investigated (N=62). GDPR lays down guidelines for the collection and processing of personal information of an organization's customers residing in the European Union. The GDPR-focused papers in our dataset mainly explored the compliance of the privacy policies of different websites with respect to GDPR principles [179]. The second most explored regional regulations were

Table 5. Classification of Data Protection Laws and Regulations Investigated in Our Dataset

Category of Regulation		Name	# Papers
Country or Region-specific		European General Data Protection Regulation (GDPR)	62
		California Consumer Privacy Act (CCPA)	3
		California Online Privacy Protection Act (CalOPPA)	2
		Singapore's Personal Data Protection Act (PDPA)	2
		European Data Protection and Retention Directive	1
		Japan Act on Protection of Personal Information (APPI)	2
		Personal Data Protection Act of the Netherlands	1
		Indian Personal Data Protection Bill	1
		Pakistan Personal Data Protection Bill	2
		Bangladesh Digital Security Act	1
		New Zealand Privacy Act	1
		South African Electronic Communications & Transactions Act (ECTA)	1
		Personal Information Protection Act of Korea	1
		China's Personal Information Protection Provision	1
Sector	Health	Health Insurance Portability and Accountability Act (HIPAA)	8
	Children	Children's Online Privacy Protection Act (COPPA)	5
	Libraries	American Library Association's Privacy Guidelines and Code of Ethics	2
		International Coalition of Library Consortia Privacy Guidelines	1
	Finance	Electronic Communications and Transition Act	2
		FDIC's Privacy Rule Handbook	1
	Education	Family Educational Rights and Privacy Act (FERPA)	1
	Mobile	GSMA's Mobile Privacy Principles	1
		Google Play's Privacy Guidelines	1
General Frameworks		Fair Information Practice Principles (FIPPs)	7
		ISO 29100:2011 Privacy Framework	3
		EU-US Safe Harbour Framework	2
		OECD Privacy Guidelines	1
		NIST SP 800-53	1
		NISTIR 8228	1

California's Consumer Privacy Act (CCPA) and **Online Privacy Protection Act (CalOPPA)** (N=5). Other countries and regions whose regulations were explored in our dataset include East Asia (Japan, China, Korea), South Asia (India, Pakistan, Bangladesh, Singapore), Netherlands, New Zealand, and South Africa.

With regards to the sector-specific regulations, the **Health Insurance Portability and Accountability Act (HIPAA)** was explored the most in our dataset (N=8). HIPAA is a federal law that protects patients' sensitive health information from being disclosed without their consent. These papers evaluated the effectiveness of this act in protecting the personal health information [119]. Other explored sector-specific regulations include those focused on children, educational institutions and libraries, mobile, and finance/banking institutions.

Among the general frameworks, the most explored framework in our dataset was the **Fair Information Practice Principles (FIPPs)** (N=7). This framework governs the collection and usage of data and manages the issues of privacy and security. The papers in our dataset explored this framework with respect to the completeness of online privacy policies [89]. Other frameworks, such as the NIST SP 800-53 guidelines, were used to create techniques, such as a semantic approach to automate the management of big data privacy policies [97].

4.4 RQ3: Sectors

To understand whether certain sectors have been investigated more than the others, we compiled all the sectors that have been targeted in the selected papers on privacy policies. Figure 7 shows a distribution of the sectors that were explored in our investigated papers.

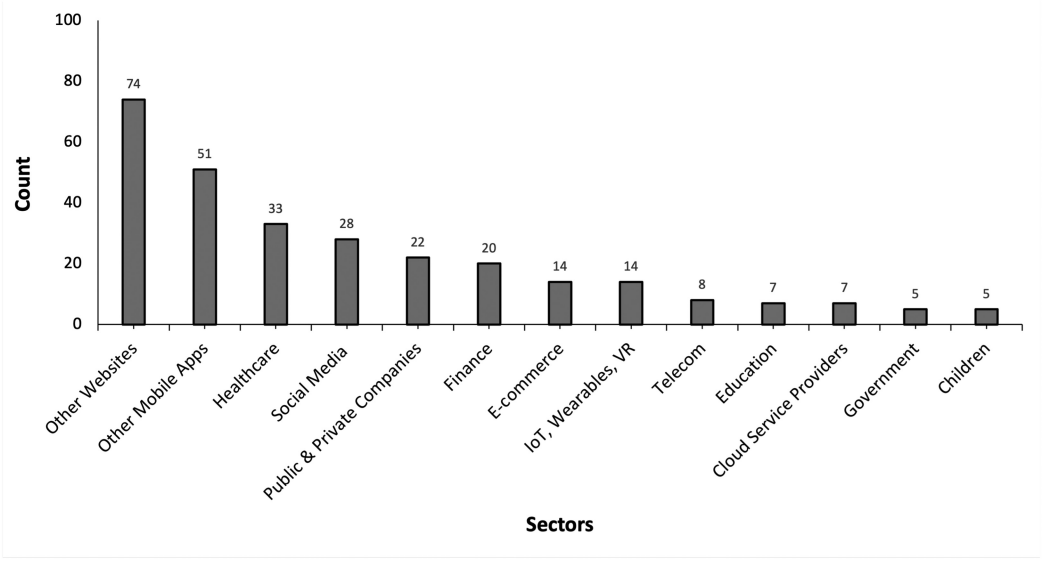


Fig. 7. Distribution of investigated sectors in our dataset.

Our analysis shows that the most explored sectors in our dataset were Social Media Platforms (N=28) [71, 171] and Healthcare (N=33) [59, 77]. Other explored sectors include Finance (N=20) [44, 102], Public and Private Companies (N=22), E-commerce (N=14) [114, 191], Cloud Service Providers (N=7) [37, 104], Education (N=7) [48, 111], Telecom (N=8) [74, 105], Government (N=5) [38, 134], and IoT (N=14) [139, 174] sectors, respectively. The least explored sectors were Search Engines [173] and Hotels [113]. Only one paper each in our dataset explored these two sectors.

Among the papers that focused on Social Media, the most explored platform was Facebook. In case of mobile apps, the most explored ones were healthcare apps. Similarly, within the Healthcare sector, topics such as mental illness, physical fitness, menstruation tracking, pregnancy monitoring, and so on, have been explored. In the E-commerce sector, various online buying and selling platforms as well as websites of physical trading stores have been explored. The Finance sector mainly included banks and other mobile money services. The Telecom sector included the Internet and telecommunication companies. The Government sector was mainly composed of the different government services and municipal websites. For the IoT sector, smart devices, their privacy concerns, as well as the compliance of their privacy policies have been particularly discussed. Some papers focused on different IoT device manufacturers and platforms. The most talked-about Search Engine was Google and for the Hotels sector, the only discussed theme revolved around the policy statements on the hotel websites and their associated privacy concerns.

Several papers (N=74, N=51) investigated websites and mobile apps, respectively, from multiple sectors. Their dataset comprised top, most popular/downloaded, or websites or apps from a certain geographical location.

Table 6 shows the types of analysis conducted across sectors. It can be seen that the Healthcare, Social Media, and Cloud Storage/Service Platforms have been investigated extensively. All aspects have been investigated in these sectors, such as policy readability, policy completeness, mismatch between policy and practice, frameworks to help write and understand policies in that specific domain, as well as user studies regarding perceptions and concerns about policies, and usability testing of proposed tools.

Table 6. Types of Analysis Conducted across Sectors

Sector	Readability Analysis	Completeness Check	Mismatch Detection	User studies and Tool/Frameworks
Healthcare	7	11	1	14
E-commerce	0	1	0	13
Social media	3	3	1	21
Cloud service platforms	1	2	2	2
Education	1	2	0	4
Finance	2	5	0	13
Telecom	1	1	0	6
Government	0	1	1	3
Children	1	0	0	4
IoT, wearables, VR	0	7	2	5
Public and private companies	3	9	0	10
Other top/popular mobile apps	7	8	15	21
Other top/popular websites	9	22	0	43

5 Discussion

The presence of a large number of articles focused on privacy policies clearly indicates that research on this topic has received a substantial amount of interest over the past years. This review focuses on the approaches used to analyze privacy policies both from the end-user perspective and the compliance aspect, as well as the various sectors and laws/regulations that have been taken into consideration. Below, we discuss the identified gaps in existing literature, followed by the implications of our findings for research and practice.

5.1 Gaps in Existing Literature

We identified several gaps in the existing literature on privacy policies that point to opportunities for future research. Below, we discuss them in detail.

- **Privacy Policy Violation Detection:** A large percentage of the existing research focuses on whether an organization's privacy policy is accessible or how compliant/complete the content of this policy is with regards to data protection principles. However, less work has been done on the investigation of actual data practices of the organization. Our dataset only contained 10% papers on this topic (see Table 3). Part of the challenge is that the data is inaccessible or is encrypted when transmitted. The existing work in this regard is mostly focused on the privacy leaks via Android applications through the use of static/dynamic analysis tools such as FlowDroid. More studies are needed to investigate the data collection practices of websites to detect their privacy policy violations.
- **User Study Demographics:** Although a considerable effort has been made towards understanding user concerns, behaviors, and perceptions related to privacy policies via quantitative and qualitative user studies (27.2% papers), there is a lack of diversity in the investigated population. We found that a majority of the recruited participants were from European, USA and Canadian, and East Asian countries. However, Middle Eastern, South Asian, and African population is relatively understudied (see Appendix B). Although crowdsourcing was utilized in online survey-based studies, in many lab/experimental studies, the participants were students. Therefore, the sample in such studies might not be a representative of the general population.
- **Policy Accessibility Investigation:** The articles investigating privacy policy accessibility show that privacy policy on mobile apps have a lower accessibility as compared to the ones

on websites [44, 79, 137, 158, 160]. However, not all categories of apps and websites have been investigated.

— **Policy Readability Improvement:** We observed that the readability of most privacy policies except for some healthcare apps has remained low until December 2023, as majority of the policies need more than eight years of education based on the Flesch-Kincaid Grade Level (see Table 1). The FKGL score for Telecom companies and other high-traffic websites is greater than 14, which means that they require more than 14 years of education to understand the policies [60, 94, 102, 164]. Similarly, the policies seem to have increased in length over time, especially post GDPR and CCPA enforcement [186]. Various summarization and visualization tools have been proposed to address this issue. However, our analysis shows that the effectiveness of only a few of these tools has been investigated via a user study [180], or their accuracy is limited when tested on a policy dataset different from that of the authors [30]. Others have either not released their tool's working prototype to the public or it does not work completely[2] [199]. Although some NLP-based Q/A tools have been proposed [90] (Table 2), future work can focus on implementing the proposed visualization/summarization/Question Answer frameworks and test their usability.

— **Uninvestigated Sectors:** Majority of the papers in our dataset focused on privacy policies related to Social Media, Popular Mobile Apps, Healthcare, E-commerce, and Finance. However, other sectors, which are equally important but have been underexplored, include Educational Institutions and Libraries, Telecom, Government, IoT and Cloud, and Children's Websites/Apps (see Table 6). Future research can focus on websites and apps in these sectors.

— **Uninvestigated Data Protection Laws:** Not surprisingly, the most investigated region-specific data protection regulation in our dataset is the GDPR (see Table 5). Other region-specific laws that were explored were from European, USA, South Asian and East Asian countries, New Zealand, and South Africa. However, there are a total of 162 countries that have enacted data privacy laws, and another 20 or more have bills or drafts of laws to do so [87]. Recent ones include the UAE Data Protection Law [43] and the Tanzania Personal Data Protection Act [7]. Therefore, it is important to investigate these and compare the findings to others. Similarly, there are several state-specific laws in the USA, such as Illinois's BIPA, that have not been investigated [6]. In terms of the sector-specific regulations, HIPAA has been widely investigated. Other unexplored sector-specific regulations include Video Privacy Protection Act, Driver Privacy Protection Act, Gramm Leach Bliley Act, Telephone Consumer Protection Act, and so on [6].

— **Distribution of Privacy Policy Datasets:** The availability of privacy policy datasets is extremely important not only to confirm the findings of the existing studies but also to help in advancing research in this field and make privacy policies more accessible to researchers. A majority of the investigated papers focused on assessing policy readability, compliance/completeness, summarization, or mismatch detection and have compiled a dataset of privacy policies in a specific sector(s). However, only a few of them have shared a link to access their datasets [93, 188]. Even among these, some of the dataset URLs do not work anymore [184]. Some of the investigated papers have utilized pre-existing public privacy policy datasets in their analysis. These include the OPP-115 [188], PrivaSeer [172], ACL/COLING [115], Ramanath et al. [11], and Zaeem et al. [133] for websites, APP-350 and MAPS [208] for Android apps, and APPC-451K [18] for iOS apps.

[2]Privacy Check: https://chrome.google.com/webstore/detail/privacycheck/poobeppenopkcbjejfjenbiepifcbclg

5.2 Implications for Research and Practice

This review contributes to both research and practice by identifying trends, practices, and opportunities for future work.

In terms of research implications, our analysis confirms findings from existing literature reviews on privacy policies and shows the importance and need for more research on privacy policies. For instance, a previous review shows low readability scores of privacy policies over the years as well as increased data collection and sharing of sensitive information [186] (see Tables 1 and 3). Similarly, the privacy policy visualization tools analyzed in another literature review [34] were completely different from the ones found in our dataset (see Table 2). However, they were mainly based on ML/NLP algorithms similar to the ones used in our dataset. The papers reviewed in the other literature reviews mainly made use of the NLP techniques for the automated analysis of privacy policies [64], whereas the papers that we reviewed used a combination of Machine Learning, Deep Learning, and NLP techniques for automated analysis (see Table 4). In terms of future research, our analysis reveals the specific sectors and regional laws/regulations that have not been fully explored yet. Similarly, we discovered the analysis methods that need more attention from the researchers. Future work can investigate these methods and look more into the effectiveness and accuracy of the existing approaches, along with studying a diverse user population to understand user concerns, behaviors, and comprehension of privacy policies. Another important need identified in this review is for the distribution of privacy policy datasets (both annotated or otherwise) to be used by others for reproducibility of their findings, as well as to leverage them in their own research.

Our research has several practical implications. The identified gaps can help organizations achieve better end-user privacy and minimize the probability of lawsuits. For instance, we have identified the types/sectors of policies and their corresponding readability scores, completeness scores, as well as approaches for finding violations between statement and practice. This information can be utilized to find out how to improve the readability and completeness with regards to the applicable privacy laws and regulations. Appendix A shows that websites' and apps' privacy policies from multiple sectors lack completeness on data handling, data retention and portability, data anonymity, sharing personal data with data aggregators, transparency, and information regarding notification of data breach. Similarly, Table 3 shows that data collection and sharing practices for most apps do not align with the privacy statement. Therefore, companies developing websites and mobile apps need to ensure the availability as well as ease of access of privacy policy and that their data practices align with the content of their privacy policies. They can also leverage the identified policy summarization/visualization/interactive tools and the readability metrics to design an effective privacy policy. Moreover, technical measures can be implemented on the user-side to automatically visualize or classify privacy practices, match them against user preferences, block unwanted data collection, and utilize privacy standards such as **Platform for Privacy Preferences (P3P)**. Additionally, regulatory measures should be established to mandate specific formats and locations for privacy policies and respect for specific privacy standards such as P3P and the **Do Not Track (DNT)** header. Similarly, measurement approaches should be designed for the users to verify compliance of organizations' policies with their actual data flows [186].

6 Conclusion

In this article, we have analyzed 202 peer-reviewed articles on privacy policies published till 31st December, 2023. We have summarized the analysis methods used to investigate privacy policies as well as the various sectors and data protection laws/regulations explored. Specifically, we found that more work is needed on evaluating the mismatch between organizations' data practices and their policy text, as well as policy readability in general, and accessibility improvement in apps.

The accuracy of the proposed automated analysis incorporating machine learning and natural language processing can be further improved, as they have been tested on a specific dataset (for instance, websites in the US) that does not perform well on the privacy policy of websites from other countries. Other gaps include the lack of public and diverse datasets, lack of diverse population in user studies and experiments, and several uninvestigated sectors and data protection laws.

Appendices

A Table: Policy Compliance/completeness Focused Papers in Our Dataset

Description of Policy Compliance/Completeness-focused Papers in Our Dataset

Type of Policies	Privacy Law/Regulation	ML/NLP	Findings	Reference
50 Banks & 54 Mobile Money Services in US	GSMA's Mobile Privacy Principles & FDIC's Privacy Rule Handbook	No	Mobile money policies are less complete than US bank policies	Bowers et al. 2017 [44]
Top 5 Digital Content Vendors of American Public Libraries	American Library Association's Code of Ethics	No	Low compliance	Lambert et al. 2015 [111]
284 South Asian websites in 10 Sectors	GDPR, Indian personal data protection data, bill, Pakistan personal data protection bill, digital security act	No	Low compliance for protection of children's data retention & portability	Javed et al. 2020 [93]
35 Online Health Social Networks	Fair Information Practice Principles	No	Only 9 websites fully complied	Savla et al. 2012 [164]
15 Social Networks for Kidney Disease	NA	No	Controls for individual privacy are missing	Costello et al. 2015 [59]
60 Homecare Companies in Ireland	GDPR	No	High level of non-compliance	Reynolds et al. 2022 [157]
11 Female Health apps	HIPAA	No	Personal data is shared with data aggregators	Erickson et al. 2022 [70]
14 South African Banks	Electronic Communications & Transactions (ECT) Act	No	No bank showed 100% compliance	Kabanda et al. 2010 [102]
Top 100 New Zealand Companies	New Zealand privacy act	No	Incomplete privacy policies	Tjhin et al. 2016 [182]
Microsoft's Health Vault & Google's GoogleHealth	HIPAA	No	Lack of standardization and transparency	Martino et al. 2010 [119]
3 Crowdsourcing websites	GDPR	No	Anonymity is not ensured	Diamantopoulou 2018 [65]
24 E-government services in West Africa	ISO/IEC 29100:2011 Principles	No	Majority are not compliant	Nwaeze et al. 2017 [134]
3 Chinese websites	China's Personal Information Protection Provision	No	Policies offer more notice than choice	Fu et al. 2019 [80]

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Type of Policies	Privacy Law/Regulation	ML/NLP	Findings	Reference
18 Major Korean, US, UK websites	GDPR and Personal Information Protection Act of Korea	No	Korean, UK websites are more compliant than US websites	Woo et al. 2020 [190]
31 Cancer Android apps	GDPR	No	39% apps had low compliance	Benjumea et al. 2020 [39]
50 websites from 4 sectors	NA	No	39.4% contained insufficient information	Pollach et al. 2007 [148]
Fortune 50, Forbes 50 Companies	FTC FIPPS	No	Public companies have better compliance than private	Peslak et al. 2005 [146]
100K websites	EU Data Protection Directive & US Safe Harbor Framework	No	Websites do not claim to follow all the requirements	Reay et al. 2009 [154]
10 Automated Privacy Policy Generators	GDPR	No	Generated policies could be incomplete/inaccurate	Sun et al. 2020 [175]
319 iOS & 69 Android Mental Health apps	NA	No	Concerns about consent, transparency, & data sharing	Robillard et al. 2019 [158]
229 E-commerce Firms in China, Taiwan, Hong Kong	Generally accepted privacy standards	No	Chinese firms have less compliance	Li et al. 2018 [114]
19 Pregnancy Monitoring Mobile apps	HIPPA and GDPR	No	No app fully complied	Bachiri et al. 2018 [33]
11 IoT Manufacturers	GDPR	No	50% do not have an adequate privacy policy	Subahi et al. 2018 [174]
72 iOS apps on Dementia	NA	No	Lack clarity on how individual-user data is handled	Rosenfeld et al. 2017 [160]
306 Malaysian Companies from 12 Sectors	Personal Data Protection Act	No	Private sector has more compliance than Government sector	Chua et al. 2017 [57]
LinkedIn	ISO 29100:2011 standard	No	Serious mismatches	Michota et al. 2015 [124]
101 Companies Offering Location/mobility Service	FTC FIPPs	No	Overall lack of completeness	Cottrill et al. 2014 [60]
24 Free Personal Health Record apps	HIPAA	No	Low compliance	Zapata et al. 2014 [201]
10 Major websites	NA	No	Low score on user data handling	Miller et al. 2012 [125]
50 Popular Free Health/Fitness Android apps	GDPR	No	Lack of completeness on 4 principles	Yaqub et al. 2023 [195]
15 Cloud Storage Android apps	NA	No	Policies lacked information of data retention period	Baalous et al. 2018 [32]

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Type of Policies	Privacy Law/Regulation	ML/NLP	Findings	Reference
IoT Manufacturers	GDPR	No	Only 10% contain notification in case of data breach	Kuznetsov et al. 2022 [110]
5 Express Chinese Companies	FTC FIPPs	No	Overall score of information security control is poor	Zhang et al. 2022 [205]
20 Free Personal Health Record websites	HIPAA	No	25% websites comply fully or partially	Carrion et al. 2011 [51]
27 Library Vendor websites	FTC FIPPs, ICOLC's privacy guidelines, ALA privacy guidelines	No	Low compliance with library profession standards and little support for the library Code of Ethics	Magi et al. 2010 [117]
Websites in OPP-115 Dataset	GDPR	Yes	NA	Alshamsan et al. 2022 [22]
70 5G Networks in UK	GDPR	Yes	51% companies complied	Zhang et al. 2023 [203]
120 Pakistani websites	GDPR & Pakistan Data Protection Act	Yes	Model showed high accuracy in computing compliance	Javed et al. 2021 [30]
2,000 Online websites	NA	Yes	European policies use fewer ambiguous words	Kaur et al. 2018 [100]
250 Companies	GDPR	Yes	76% do not contain all of the requirements	Mueller et al. 2019 [130]
1,304 Privacy Policies of Japanese Companies	Act on Protection of Personal Info.	Yes	90% do not describe data retention and specific audience	Mori et al. 2022 [129]
100 websites from 15 Sectors	FIPPs, OECD	Yes	NA	Guntamukkala et al. 2015 [89]
3,000 Web Services	GDPR	Yes	Model showed high accuracy in computing compliance	Elluri et al. 2021 [69]
Alexa Top 5,000 websites	NA	Yes	High accuracy in segment classification	Adhikari et al. 2022 [10]
Top websites	GDPR	Yes	Policies after GDPR have better compliance	Bateni et al. 2022 [36]
170 Financial apps in China	GDPR	Yes	NA	Shen et al. 2023 [165]
8,614 Companies in European Union	GDPR	Yes	Large companies have better compliance than small ones	Aberkane et al. 2022 [9]
100 Popular Chinese apps	Personal Information Protection Law	Yes	Proposed model can assist in automated analysis	Tang et al. 2023 [178]
224 Cloud Services	GDPR	Yes	Proposed model can assist in automated analysis	Razavisousan et al. 2021 [153]
Android Apps	GDPR	Yes	Generates GDPR-compliant permission policy statements	Mcconkey et al. 2023 [120]

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Type of Policies	Privacy Law/Regulation	ML/NLP	Findings	Reference
Car Makers Mobile apps	GDPR & CCPA	Yes	Insufficient information on data management	Bodei et al. 2023 [41]
6 Wearable Device Manufacturers	NISTIR 8228	Yes	Hexoskin has the lowest compliance	Echenim et al. 2023 [67]
1726 Virtual Reality apps	GDPR	Yes	48.1% VR apps violate the data minimization principle	Zhan et al. 2023 [202]
54 Policies from Different Sectors	GDPR	Yes	Model has 84.6% accuracy in detecting violation	Cejas et al. 2023 [55]
9,000 Android apps	GDPR	Yes	80% apps fail to meet transparency requirement	Rodriguez et al. 2023 [159]
234 Funding Companies in Luxembourg	GDPR	Yes	Model has 92.9% precision in detecting violation	Amaral et al. 2022 [24]
234 Fund Management Companies in Luxembourg	GDPR	Yes	85% precision in detecting incompleteness issues	Torre et al. 2020 [184]
Most Downloaded Android apps	Google Play privacy guidelines	Yes	1% apps fully comply with Google Play privacy guidelines	Verderame et al. 2020 [185]
5829 Amazon Alexa Skills	GDPR	Yes	Majority of skills suffer from policy non-compliance	Xie et al. 2022 [193]
187 Third-party Library & 641 Host apps	GDPR and CCPA	Yes	More than 50% violate data usage & sharing requirements	Zhao et al. 2023 [207]
1,562 Free & Premium Content websites	GDPR	Yes	Premium websites are more compliant than free websites	Alabduljabbar et al. 2022 [17]
304 Android apps	GDPR	Yes	Proposed automated analysis approach has 90% accuracy	Liu et al. 2021 [116]
50K Policies from 1996–2021	GDPR & CCPA	Yes	Incomplete on several principles such as user rights	Wagner et al. 2022 [186]
1,080 websites	GDPR	Yes	68% failed to comply with at least one requirement	Rahat et al. 2022 [151]
Top 100 Fortune Companies, 67 Rogue websites	EU-US Safe Harbour framework	Yes	High compliance by companies, Low compliance by rogue websites	Boldt et al. 2019 [42]
110 EU+Non-EU Mobile Payment Systems	GDPR	Yes	Low compliance index	Akanfe et al. 2020 [13]

B Table: User Perceptions, Behavior, and Comprehension of Privacy Policies

Description of Privacy Policy Focused User Studies in Our Dataset

Design	Goal	Sample Size	Demographics	Reference
Lab experiment	Use of eye-gaze tracking to see how participants read policies based on the presentation format	128	Undergraduate students at a university in Israel	Steinfeld 2016 [173]
Lab experiment	Readability of privacy in mobile vs. desktop environment	50	-	Singh et al. 2011 [167]
Lab experiment	Pre/post test questionnaire/interviews to study the effectiveness of policy summarization tool	15	German	Ortloff et al. 2020 [141]
Lab experiment	Evaluate subjects' ability to comprehend the content of privacy policies	50	-	Singh et al. 2011 [168]
Lab experiment	Pre/post test questionnaire to study whether visualized version of privacy policy leads to greater awareness	120	Undergraduate students at universities in Greece	Soumelidou et al. 2019 [171]
Lab experiment	Measure the effectivess of PPmark tool in understanding privacy policies	40	Undergraduate students from university in Brazil	Pontes et al. 2017 [150]
Lab experiment (between subject)	Measure the effect of privacy policy visualizations on trust	128	IS and economics Students in Germany	Becker et al. 2014 [37]
Lab experiment	Use of control/restricted theory and cognitive theory of multimedia learning, to demonstrate how perceived control and learning can affect trust on text vs. video format of privacy policy	30	International graduate students	Lee et al. 2013 [113]
Lab experiment	Evaluate the effectiveness of policy visualization	90	Students	Reinhardt et al. 2021 [156]
Lab experiment	Evaluate the effectiveness of ontology on reading time	5	-	Kaur et al. 2022 [99]
Lab experiment	Test the effectiveness of policy annotation tool in reducing its reading time	20	-	Alabduljabbar et al. 2021 [15, 16]
Lab experiment	Evaluate the usability of policy parsing tool	26	Users in companies	Brodie et al. 2006 [47]
Lab experiment	Evaluate the usability of autocompliance web tool	10	IT managers and students	Liu et al. 2021 [116]

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Design	Goal	Sample Size	Demographics	Reference
Lab experiment	Test the effectiveness of policy authoring tools	36	IT company employees	Karat et al. 2006 [98]
Online experiment (between subject)	Analyze user trust based on the presentation format of privacy policy	717	MTurkers in Canada, US	Aimeur et al. 2016 [12]
Online experiment (between subject)	Measure perceived readability of privacy policy vs. trust in a provider	539	German	Ermakova et al. 2016 [72]
Online experiment (between subject)	Impact of adoption of GDPR and readable privacy policy on customers' trust and intention to disclose information	255	MTurkers in US	Zhang et al. 2019 [206]
Online experiment (between subject)	Understand comprehension of privacy policy in text, video, & usable format	208	Online users in Ireland	Faurie et al. 2020 [78]
Online experiment (between subject)	Effect of policy summary and confidence rating on trust	944	Japanese	Bracamonte et al. 2020 [46]
Online experiment (between subject)	Effectiveness of policy summarization tool in communicating risk and generating user interest in policy content	1040	Japanese	Bracamonte et al. 2019 [45]
Online experiment (between subject)	Effectiveness of policy visualization tool	65	Students	Brunotte et al. 2022 [49]
Online experiment (between subject)	Effectiveness of various policy visualization formats	764	MTurk participants	Kelley et al. 2010 [101]
Online experiment (between subject)	Effectiveness of various policy visualization formats	749	Internet users	McDonald et al. 2009 [122]
Online experiment	Effectiveness of PriBot tool in answering questions about policy content	700	MTurk participants	Harkous et al. 2018 [90]
Online, Lab experiment	Effectiveness of visualization on policy comprehension	520 & 12	Internet users & university	Reeder et al. 2008 [155]
Online survey	Measure perceived readability of privacy policy vs. trust in a provider			
using Partial Least Squares (PLS)		440	German	Ermakova et al. 2014 [71]

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Design	Goal	Sample Size	Demographics	Reference
Online survey	User attitudes towards privacy policies using Online Privacy Literacy Scale	500	Seniors	Oeldorf et al. 2019 [136]
Online survey	Perceptions about privacy policy	356	Online users	Altamimi et al. 2018 [23]
Online survey	Perceptions and awareness about privacy policy	366	Students	Talib et al. 2014 [176]
Online survey	Effect of policy length on time spent reading & understanding (using SEM)	330	Online users	Meier et al. 2020 [123]
Online survey	Understand user awareness of Snapchat privacy policy and its effect on their discontinuous usage intention	201	Saudia Arabians	Al-Saqer et al. 2018 [14]
Online survey	Understand user perceptions and behavior regarding privacy settings and privacy policies	1,391	German	Rudolph et al. 2018 [161]
Online survey	Explore whether users read terms of use and privacy policies	170	College students	Moallem et al. 2018 [127]
Online survey	Explore factors (using SEM), which may inhibit nurses' intentions to violate privacy policy concerning medical records	262	Nurses	Kuo et al. 2017 [109]
Online survey	Explore factors that motivate hospital's IT staff's compliance with medical records privacy policy	310	IT Staff of 7 medical centers in Taiwan	Sher et al. 2017 [166]
Online survey	User concerns about privacy policies on social network & financial websites	323	MTurkers in US	Alduaij et al. 2019 [19]
Online survey	Investigate how risk perception affects relationship between changes in privacy policies and users' willingness to share information	1,116	-	Gerlach et al. 2015 [84]
Online survey	Effect of familiarity with social network and presence of Privacy Seals on how users process information in Social media privacy policies	167	Students	Yang et al. 2015 [194]
Online survey	Understand user expectations of informed consent in privacy policies	8,621	EU users	Custers et al. 2014 [61]

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Design	Goal	Sample Size	Demographics	Reference
Online survey	Investigate the effectiveness of privacy policies and privacy seals security icons on corporate trustworthiness and reputation management	600	Japanese university students	Orito et al. 2013 [140]
Online survey	Identify the relationships between each FTC FIPPs principle in privacy statements of websites and user trust/privacy concern	500	Taiwan and Russia	Wu et al. 2012 [191]
Online survey	Analyze the effect of policy design elements (length, visibility, specificity) on its perceived importance & relevance	-	-	Capistrano et al. 2015 [50]
Online survey	Identify motivators and blockers that influence users' willingness to read privacy policies	655	MTurk Participants	Ibdah et al. 2021 [91]
Online survey	Investigate if users understand the policy without automated tools	102	MTurkers in US and EU	Rahat et al. 2022 [151]
Online survey	Impact of perception of company's respect for customer privacy on customers' trust in the company	269	Online users	Lauer et al. 2006 [112]
Survey	Analyze the effect of neutralization theory on privacy policy violation	66	Saudi Medical students	Altamimi et al. 2018 [23]
Survey	Investigate how WhatsApp's privacy policy update affected users' behavior and attitude towards WhatsApp and other messaging apps	1,525	WhatsApp users in Spain, UK, Mexico, South Africa	Griggio et al. 2022 [88]
Focus group	Exploration of learner experiences with privacy policies	3	Convenience sampling	Pandya et al. 2022 [142]
Interviews	User perceptions about different presentation formats of privacy policies	5	US	Pierce et al. 2018 [147]
Interviews	User perceptions about privacy policies and awareness about ECT act	13	South African bank users	Keshav et al. 2010 [102]
Semi-structured interviews	Explore factors that influence users' intention to read privacy statements	223	Netherlands employees	Beldad et al. 2010 [38]
Heuristic evaluation	Usability of Privacy Policy Visualization using STARP evaluation framework	-	-	Ghazinour et al. 2016 [85]

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